

B. Tech. Project AY 2026-27

CA 1 Presentation

Project ID: _____

Title of the Project: _____



**SYMBIOSIS INSTITUTE OF
TECHNOLOGY, PUNE**

Name of the Guide: _____

Name of the Industry Mentor: _____

Group Members:

1. Name (PRN)
2. Name (PRN)
3. Name (PRN)
4. Name (PRN)

Scope evolution and reporting window

Started as a pitch; became a measured recovery study for gfx90c on Fedora 43.

The evidence chain now runs through spoofing, benchmarks, and source builds.

The same workflow is being extended to gfx902 and later upstream work.

Progress reports: 01-07-26 to 15-07-26, then 15-07-26 to 31-07-26.

Source sequence: blogs #1 to #8, with #1 and #2 preceding the July window.

Timeline placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Introduction

Project title: **Generalisable Pipeline for Backporting ROCm to Obsolete AMD Hardware.**

Working context: FOSS United Foundation, Season of Commits.

Core case study: gfx90c on a Ryzen 5650U APU, with gfx902 as the next validation target.

Main shift: from hardware speculation to evidence-backed software recovery.

Native setup screenshot

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Problem statement

Fedora can install ROCm natively, but higher-level packages still fail on gfx90c. PyTorch and rocm-examples miss required gfx90c artifacts. GPU spoofing helps validation, but it does not match the real silicon path.

The software stack is fragmented across system ROCm, packaged ROCm, and source builds.

Failure surface placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Objectives of the project

Restore gfx90c support in a repeatable way.

Measure each recovery stage against a native baseline.

Port the same workflow to gfx902.

Keep the path general enough for future dropped AMD targets.

Objective flowchart placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Project plan with timeline

6 months: complete gfx90c port recovery.

9 months: finish an end-to-end CI/CD pipeline.

Next project: contribute directly to Mojo, LLVM, ROCm, PyTorch, and vLLM.

Trigger: a SOTA RDNA 4 GPU from FOSS United Foundation.

Gantt timeline placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Literature review - phase 1

Blog #1 sets the motivation and the gfx90c / Vega 7 target.

Blog #2 shows native ROCm 6.4.4 on Fedora 43 and the first PyTorch failure.

The first progress report compresses these findings into a working baseline.

Post sequence placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Literature review - phase 2

Blog #3 tests GPU spoofing via an HSA override.

Blog #4 adds AMD first-party benchmarks and FHS packaging context.

The first progress report turns this into benchmark-backed evidence.

Benchmark comparison placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Literature review - phase 3

Blog #5 builds GPU programming and execution-model context.

Blog #6 studies rocm-examples failures on gfx90c.

Blogs #7 and #8 move into source builds and the Vega execution / memory model.

Assembly and ISA placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Gap in the research, technology, and methodology

There was no single reproducible recovery workflow.

Packaged binaries omitted target-specific gfx90c data.

Existing fixes were scattered across posts, issues, and commands.

The missing layer was not just support, but a measurable method.

Sankey or gap map placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Description of the proposed solution

Use a staged recovery pipeline instead of a single workaround.

Move from native validation to spoofing, mixed stacks, containers, and source builds.

Treat each step as a separate diagnostic layer.

Reuse the same workflow for gfx902 and later targets.

Pipeline diagram placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Requirement analysis

Fedora 43 host with ROCm 6.4.4.

HIP, rocBLAS, PyTorch, and rocm-examples coverage.

A benchmark set that includes bandwidth, GEMM, and runtime checks.

A second device family for cross-validation.

Requirements matrix placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Technology stack

Fedora 43, ROCm 6.4.4, HIP, HIPCC.

LLVM, clang, rocm-examples, rocBLAS, PyTorch.

CLBlast, clpeak, Geekbench, and AMD benchmark clients.

Bash, CMake, Git, and source-level patching.

Layered stack diagram placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Design

Design follows a bottom-up recovery model.

Each layer isolates packaging, build, runtime, or hardware limits.

The same design should explain success on gfx90c and failure on missing paths.

Architecture block diagram placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Development and implementation - baseline

Verified native ROCm on
Fedora 43.

Confirmed HIP execution on
gfx90c.

Traced the first PyTorch failure
to missing rocBLAS kernels.

Established spoofing as a
useful but imperfect baseline.

Command output placeholder

Insert figure, screenshot,
chart, or diagram here.

Why it helps: anchors the
slide in evidence and keeps
the text sparse.

Development and implementation - source build path

Built rocm-examples from
source against ROCm 6.4.4.

Reached the missing
`main_gfx90c.s` barrier.

Ported
`assembly_to_executable`
and related build pieces.

Studied how `.hip`, LLVM IR, and
AMDGPU code fit together.

Source build / assembly screenshot placeholder

Insert figure, screenshot,
chart, or diagram here.

Why it helps: anchors the
slide in evidence and keeps
the text sparse.

Testing and debugging - failure analysis

Classified failures into
packaging, metadata, missing
Tensile data, and
hardware-limited cases.

Compared native, spoofed, and
source-built behaviour.

Kept benchmark runs attached
to each recovery step.

Error taxonomy / Sankey placeholder

Insert figure, screenshot,
chart, or diagram here.

Why it helps: anchors the
slide in evidence and keeps
the text sparse.

Testing and debugging - benchmark comparison

Benchmarks covered bandwidth, GEMM, clpeak, Geekbench, and rocBLAS.

Spoofing stayed close on some metrics and diverged on others.

The source-build path clarified which failures were structural.

Native vs spoofed graph placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Project outcome

gfx90c is now a studied, testable target rather than an unknown.

The runtime path works for basic HIP code.

The remaining blockers sit in packaging, kernels, and generated artifacts.

gfx902 is ready for the same validation sequence.

Outcome dashboard placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

Conclusion

The project moved from a broad idea to a reproducible recovery method.

The strongest evidence points to software and packaging gaps.

The next stage is to finish the gfx90c port and then harden the pipeline.

The later roadmap expands into direct upstream work after new hardware arrives.

Roadmap end-state placeholder

Insert figure, screenshot, chart, or diagram here.

Why it helps: anchors the slide in evidence and keeps the text sparse.

References

AMD vis-à-vis GPUs #1 - #8.

Progress Report No #1 and Progress Report No #2.

AMD ROCm documentation, LLVM source history, and the Vega ISA reference guide.

Fedora packaging notes and issue threads related to gfx90c support.